

Research Methods — Module 02: Agents — Study Questions

1. What is computational phenotyping? How does it differ from behavioral phenotyping?
2. Why might two people with identical task accuracy have different computational phenotypes?
3. List the Active Inference parameters that constitute a computational phenotype. What does each measure?
4. What task features help dissociate model parameters?
5. Describe the model fitting procedure for behavioral data. What optimization methods are used?
6. What is hierarchical fitting (PEB)? Why is it preferable to individual fitting in group studies?
7. What is parameter recovery? How do you perform it?
8. What constitutes "good" parameter recovery? What metrics are used?
9. What is parameter identifiability? How do you test for it?
10. What does it mean if two parameters are highly correlated in recovery?
11. How does model comparison on simulated data validate model distinguishability?
12. What is a transdiagnostic approach to psychiatry? How does computational phenotyping enable it?
13. How might computational parameters guide treatment selection?
14. What is normative modeling? How does it differ from group comparison?
15. What is longitudinal computational monitoring? What changes in parameters indicate treatment response?
16. What sample sizes are needed for reliable computational phenotyping?
17. What software tools are available for fitting Active Inference models?
18. What should be reported in a computational phenotyping paper?
19. What are the ethical implications of computational phenotyping in psychiatry?
20. Critically evaluate: Is computational phenotyping ready for clinical use, or still purely a research tool?